The diurnal rhythm is an elementary source of our idea of time: night follows day; day follows night. We count the beats of this great clock: we count the days. In the ancient consciousness of humanity, time is, above all, this counting of days.

As well as the days, we then count the years and the seasons, the cycles of the moon, the swings of a pendulum, the number of times that an hourglass is turned. This is the way in which we have traditionally conceived of time: counting the ways in which things change.

Aristotle is the first we are aware of to have asked himself the question "What is time?," and he came to the following conclusion: time is the measurement of change. Things change continually. We call "time" the measurement, the counting of this change.

Aristotle's idea is sound: time is what we refer to when we ask "when?" "After how much *time* will you return?" means "When will you return?" The answer to the question "when?" refers to something that happens. "I'll return in three days' time" means that between departure and return the sun will have completed three circuits in the sky. It's as simple as that.

So if nothing changes, if nothing moves, does time therefore cease to pass?

Aristotle believed that it did. If nothing changes, time does not pass—because time is our way of situating ourselves in relation to the changing of things: the placing of ourselves in relation to the counting of days. Time is the measure of change:[42] if nothing changes, there is no time.

But what is then the time that I hear coursing in the silence? "If it is dark and our bodily experience is nil," Aristotle writes in his *Physics*, "but some change is happening within the mind, we immediately suppose that some time has passed as well."[43] In other words, even the time that we perceive flowing within us is the measure of a movement: a movement that is internal. . . . If nothing moves, there is no time, because time is nothing but the registering of movement.

Newton, instead, assumes the exact opposite. In his magnum opus, the *Principia*, he writes:

> I do not define Time, Space, Place and Motion, as being well known to all. Only I must observe, that the common people conceive those quantities under no other notions but from the relation they bear to sensible objects. And thence arise certain prejudices, for the removing of which, it will be convenient to distinguish them into Absolute and Relative, True and Apparent, Mathematical and Common.[44]

In other words, Newton recognizes that a kind of "time" exists that measures days and movements: the one treated by Aristotle (relative, apparent, and common). But he also contends that, in addition to this, *another*